

# Moments and Singular Perturbation

Yoonjae Hwang

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## 1 Problem Setup

**SYSTEM:**

$$\dot{x} = A_{11}x + A_{12}z + B_1u \quad (1)$$

$$\varepsilon \dot{z} = A_{21}x + A_{22}z + B_2u \quad (2)$$

$$y = C_1x \quad (3)$$

No  $z$  term as otherwise we would have feedforward in the reduced model.

In state space form:

$$\begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ \frac{1}{\varepsilon}A_{21} & \frac{1}{\varepsilon}A_{22} \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} + \begin{bmatrix} B_1 \\ \frac{1}{\varepsilon}B_2 \end{bmatrix} u \quad (4)$$

**SIGNAL:**

$$\dot{w}_1 = S_{11}w_1 + S_{12}w_2 \quad (5)$$

$$\varepsilon \dot{w}_2 = S_{21}w_1 + S_{22}w_2 \quad (6)$$

**REDUCED MODEL SIGNAL:**

$$S_{22}w_2 + S_{21}w_1 = 0 \quad (7)$$

$$w_2 = -S_{22}^{-1}S_{21}w_1 \quad (8)$$

$$\dot{w}_1 = (S_{11} - S_{12}S_{22}^{-1}S_{21})w_1 \quad (9)$$

$$u = (L_{g1} - L_{g2}S_{22}^{-1}S_{21})w_1 \quad (10)$$

**REDUCED MODEL OF THE SYSTEM:**

$$A_{22}z + A_{21}x + B_2u = 0 \quad (11)$$

$$z = -A_{22}^{-1}(A_{21}x + B_2u) \quad (12)$$

$$\dot{x} = (A_{11} - A_{12}A_{22}^{-1}A_{21})x + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21})w_1 \quad (13)$$

**FORWARD SYLVESTER EQUATION FOR THE FULL SYSTEM:**

$$\begin{bmatrix} A_{11} & A_{12} \\ \frac{1}{\varepsilon}A_{21} & \frac{1}{\varepsilon}A_{22} \end{bmatrix} \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} + \begin{bmatrix} B_1L_{g1} & B_1L_{g2} \\ \frac{1}{\varepsilon}B_2L_{g1} & \frac{1}{\varepsilon}B_2L_{g2} \end{bmatrix} = \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} \begin{bmatrix} S_{11} & S_{12} \\ \frac{1}{\varepsilon}S_{21} & \frac{1}{\varepsilon}S_{22} \end{bmatrix} \quad (14)$$

**FORWARD SYLVESTER EQUATION FOR THE REDUCED MODEL**

$$(A_{11} - A_{12}A_{22}^{-1}A_{21})\tilde{\Pi} + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21}) = \tilde{\Pi}(S_{11} - S_{12}S_{22}^{-1}S_{21}) \quad (15)$$

Can we show that as  $\varepsilon \rightarrow 0$ , the forward sylvester equation for the full system tends towards the forward sylvester equation for the reduced model?

## 2 Proof (I guess?)

Expanding out (14) we have:

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1L_{g1} = \Pi_{x1}S_{11} + \frac{1}{\varepsilon}\Pi_{x2}S_{21}, \quad (16a)$$

$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1L_{g2} = \Pi_{x1}S_{12} + \frac{1}{\varepsilon}\Pi_{x2}S_{22}, \quad (16b)$$

$$\frac{1}{\varepsilon}(A_{21}\Pi_{x1} + A_{22}\Pi_{z1}) + \frac{1}{\varepsilon}B_2L_{g1} = \Pi_{z1}S_{11} + \frac{1}{\varepsilon}\Pi_{z2}S_{21}, \quad (16c)$$

$$\frac{1}{\varepsilon}(A_{21}\Pi_{x2} + A_{22}\Pi_{z2}) + \frac{1}{\varepsilon}B_2L_{g2} = \Pi_{z1}S_{12} + \frac{1}{\varepsilon}\Pi_{z2}S_{22}. \quad (16d)$$

Multiplying (16c) and (16d) by  $\varepsilon$  we have:

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1L_{g1} = \Pi_{x1}S_{11} + \frac{1}{\varepsilon}\Pi_{x2}S_{21}, \quad (17a)$$

$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1L_{g2} = \Pi_{x1}S_{12} + \frac{1}{\varepsilon}\Pi_{x2}S_{22}, \quad (17b)$$

$$A_{21}\Pi_{x1} + A_{22}\Pi_{z1} + B_2L_{g1} = \varepsilon\Pi_{z1}S_{11} + \Pi_{z2}S_{21}, \quad (17c)$$

$$A_{21}\Pi_{x2} + A_{22}\Pi_{z2} + B_2L_{g2} = \varepsilon\Pi_{z1}S_{12} + \Pi_{z2}S_{22}. \quad (17d)$$

We now consider the  $\varepsilon$  series expansion of  $\Pi_{x1}$ ,  $\Pi_{x2}$ ,  $\Pi_{z1}$ ,  $\Pi_{z2}$ . We can see that by multiplying (17a) and (17b) by  $\varepsilon$ , and setting  $\varepsilon$  to 0 we have that the zeroth order term for  $\Pi_{x2}$  is 0.

We then write:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1L_{g1} = \Pi_{x1}^{(0)}S_{11} + \Pi_{x2}^{(1)}S_{21} + O(\varepsilon), \quad (18a)$$

$$A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} = \Pi_{x1}^{(0)}S_{12} + \Pi_{x2}^{(1)}S_{22} + O(\varepsilon), \quad (18b)$$

$$A_{21}\Pi_{x1}^{(0)} + A_{22}\Pi_{z1}^{(0)} + B_2L_{g1} = \Pi_{z2}^{(0)}S_{21} + O(\varepsilon), \quad (18c)$$

$$A_{22}\Pi_{z2}^{(0)} + B_2L_{g2} = \Pi_{z2}^{(0)}S_{22} + O(\varepsilon) \quad (18d)$$

Each of the 4 equations in (18) are essentially polynomials in epsilon.

After equating coefficients, we can write  $\Pi_{x2}^{(1)}$  as such (Assuming  $S_{22}$  is invertible):

$$\Pi_{x2}^{(1)} = \left( A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} - \Pi_{x1}^{(0)}S_{12} \right) S_{22}^{-1} \quad (19)$$

Substituting (19) into (18a) we have:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1L_{g1} = \Pi_{x1}^{(0)}S_{11} + \left( A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} - \Pi_{x1}^{(0)}S_{12} \right) S_{22}^{-1}S_{21} + O(\varepsilon) \quad (20)$$

Rearranging we have:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1L_{g1} = \Pi_{x1}^{(0)}(S_{11} - S_{12}S_{22}^{-1}S_{21}) + \left( A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} \right) S_{22}^{-1}S_{21} + O(\varepsilon) \quad (21)$$

Similarly, from (18c) we can substitute in an expression for  $\Pi_{z1}^{(0)}$ :

$$A_{11}\Pi_{x1}^{(0)} + A_{12}A_{22}^{-1} \left( \Pi_{z2}^{(0)}S_{21} - A_{21}\Pi_{x1}^{(0)} - B_2L_{g1} \right) + B_1L_{g1} = \Pi_{x1}^{(0)}(S_{11} - S_{12}S_{22}^{-1}S_{21}) + \left( A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} \right) S_{22}^{-1}S_{21} + O(\varepsilon) \quad (22)$$

Rearranging we have:

$$\begin{aligned} & (A_{11} - A_{12}A_{22}^{-1}A_{21})\Pi_{x1}^{(0)} + (B_1 - A_{12}A_{22}^{-1}B_2)L_{g1} \\ & + \left( A_{12}A_{22}^{-1}\Pi_{z2}^{(0)} - \left( A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} \right) S_{22}^{-1} \right) S_{21} \\ & = \Pi_{x1}^{(0)}(S_{11} - S_{12}S_{22}^{-1}S_{21}) + O(\varepsilon) \end{aligned} \quad (23)$$

Simplifying the middle (...)  $S_{21}$  term using (18d):

$$\left( A_{12}A_{22}^{-1}\Pi_{z2}^{(0)} - A_{12}\Pi_{z2}^{(0)}S_{22}^{-1} - B_1L_{g2}S_{22}^{-1} \right) S_{21} \quad (24)$$

$$= \left( A_{12}A_{22}^{-1} \left( \Pi_{z2}^{(0)}S_{22} - A_{22}\Pi_{z2}^{(0)} \right) S_{22} - B_1L_{g2}S_{22}^{-1} \right) S_{21} \quad (25)$$

$$= \left( A_{12}A_{22}^{-1}B_2L_{g2}S_{22}^{-1} - B_1L_{g2}S_{22}^{-1} \right) S_{21} \quad (26)$$

$$= - \left( B_1 - A_{12}A_{22}^{-1}B_2 \right) L_{g2}S_{22}^{-1}S_{21} \quad (27)$$

Substituting (27) back into (23) and factorising we have:

$$(A_{11} - A_{12}A_{22}^{-1}A_{21})\Pi_{x1}^{(0)} + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21}) \quad (28)$$

$$= \Pi_{x1}^{(0)}(S_{11} - S_{12}S_{22}^{-1}S_{21}) + O(\varepsilon) \quad (29)$$

which is of  $O(\varepsilon)$  error from (15). Thus as  $\varepsilon \rightarrow 0$ , (29) approaches (15), and the value  $\Pi_{x1} \rightarrow \Pi_{x1}^{(0)} = \tilde{\Pi}$ .

### 3 Next Steps

- Expression for  $\Pi_{x2}$  (Is it needed?)
- Higher order epsilon expansion?
- Assumptions, and rigour?